

Deepfake Technology: Advancements in Detection and Generation Methods

**Bachelor of Technology
in
Computer Science & Engineering
(Artificial Intelligence)**

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March 2026

Abstract

Deep learning has revolutionized analytics, data processing, and computer vision, delivering unprecedented capabilities in pattern recognition, image synthesis, and video manipulation. However, the same powerful techniques have also enabled the rapid proliferation of highly realistic deepfakes, posing severe threats to security, privacy, trust in digital media, and democratic processes. These synthetic media artifacts can be weaponized for misinformation campaigns, identity fraud, financial scams, political manipulation, and even malware development, thereby undermining public confidence in visual evidence and creating new challenges in digital forensics.

Recent reports indicate an explosive growth in deepfake volume, surging from approximately 500,000 shared files in 2023 to a projected 8 million by the end of 2025 — a staggering 1,500% increase. Deepfake-related fraud attempts have risen by over 3,000% in recent years, with contact center losses potentially reaching \$44.5 billion in 2025 alone, highlighting the urgent need for effective countermeasures.

While the majority of existing survey papers concentrate primarily on deepfake detection strategies, the underlying generation methodologies, tools, and their continuous evolution remain relatively underexplored. This paper presents a comprehensive and up-to-date review of both deepfake generation and detection techniques spanning 2023 to 2025. It provides an in-depth comparative analysis of state-of-the-art architectures, including GAN-based, diffusion-model-driven, and transformer-based approaches, alongside major benchmark datasets, open-source and commercial tools, and rigorous performance benchmarks.

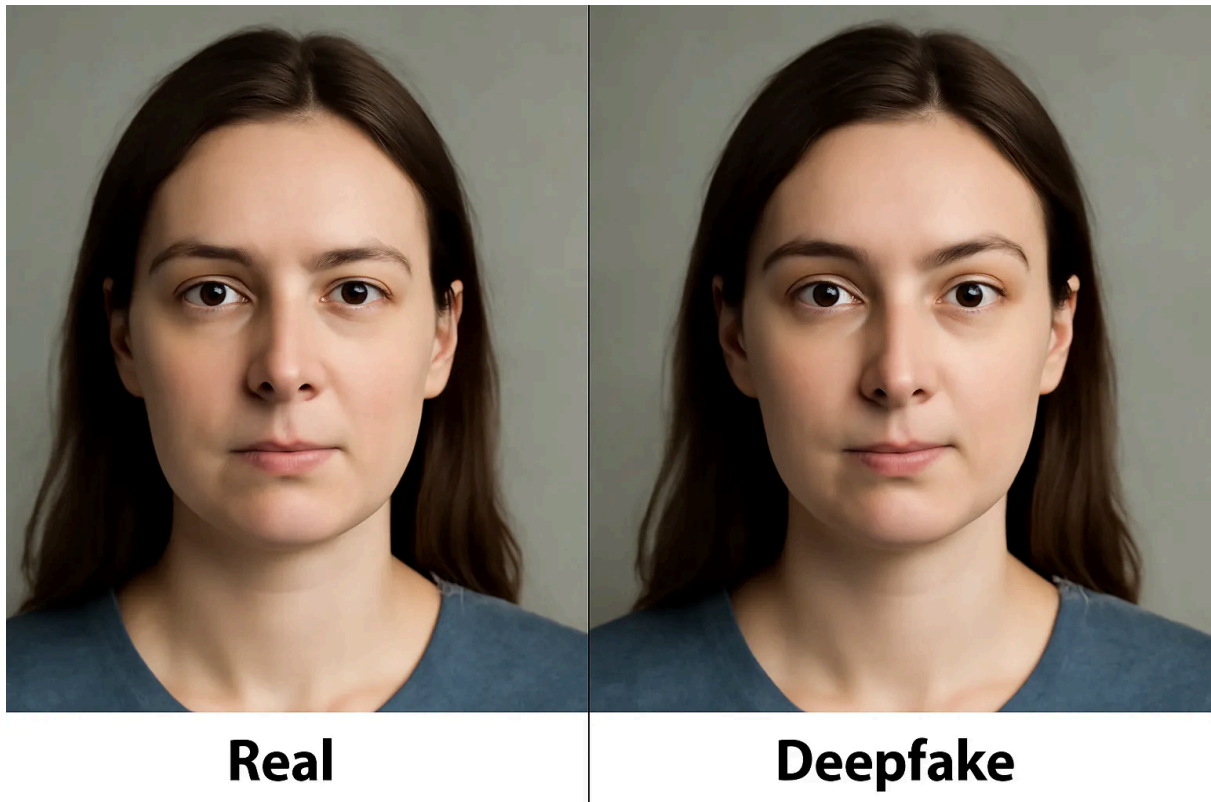
The study significantly extends the original five-page IEEE ICCCA 2025 paper by incorporating the latest post-2025 advancements, real-world impact statistics, emerging challenges such as adversarial robustness, cross-dataset generalization, and real-time deployment issues. It also highlights ethical concerns and regulatory implications associated with synthetic media. Through detailed visual examples, quantitative evaluations, and trend analysis, this review offers a holistic understanding of the deepfake ecosystem, serving as a valuable resource for researchers, practitioners, developers, and policymakers striving to mitigate the growing risks associated with synthetic media in an increasingly AI-driven world. Finally, it outlines promising future directions, including the integration of multimodal foundation models and standardized provenance frameworks, to foster more resilient media authentication systems.

1. Introduction

Deep learning has emerged as a cornerstone of modern computer vision and data analytics. Despite its transformative benefits, it has also facilitated the development of malicious technologies, including deepfakes, which threaten digital authenticity and cybersecurity.

Deepfake videos are generated by replacing or manipulating a person's identity in visual media using deep neural networks. Although extensive research has been conducted on detection mechanisms, limited attention has been given to understanding the generation pipeline in parallel.

This work expands the current research landscape by analyzing both generation and detection methodologies, supported by datasets, tools, and cross-domain evaluations. Recent developments in Generative Adversarial Networks (GANs), diffusion models, and transformer-based architectures have significantly improved realism, making detection increasingly challenging.



The rapid growth of deepfake content—projected to exceed 8 million instances by 2025—highlights the urgency for robust detection systems.

2. Deepfake Generation Methods (2023–2025)

Recent advancements in deepfake generation have shifted from traditional GAN-based models to more sophisticated diffusion-based and hybrid architectures.

Face Swapping and Re-enactment

These methods map facial features from a source to a target using landmark detection and texture synthesis, enabling realistic identity transfer.

Audio-Driven Talking Heads

Models such as Wav2Lip generate synchronized lip movements based on input audio, enabling highly realistic speech-driven animations.

Diffusion-Based Synthesis

Diffusion models (2024–2025) provide superior temporal consistency and visual fidelity compared to GANs, making generated content nearly indistinguishable from real data.

Open-source tools such as DeepFaceLab and DeepFaceLive have enabled real-time deployment of these techniques.



Despite their capabilities, these technologies raise significant ethical concerns regarding consent, misuse, and misinformation.

3. Deepfake Detection Methods (2023–2025)

Detection techniques have evolved from traditional CNN-based approaches to multi-modal and transformer-based systems.

Spatial Domain Methods

CNN architectures (e.g., Xception, ResNet) identify visual inconsistencies such as blending artifacts and unnatural textures.

Temporal Domain Methods

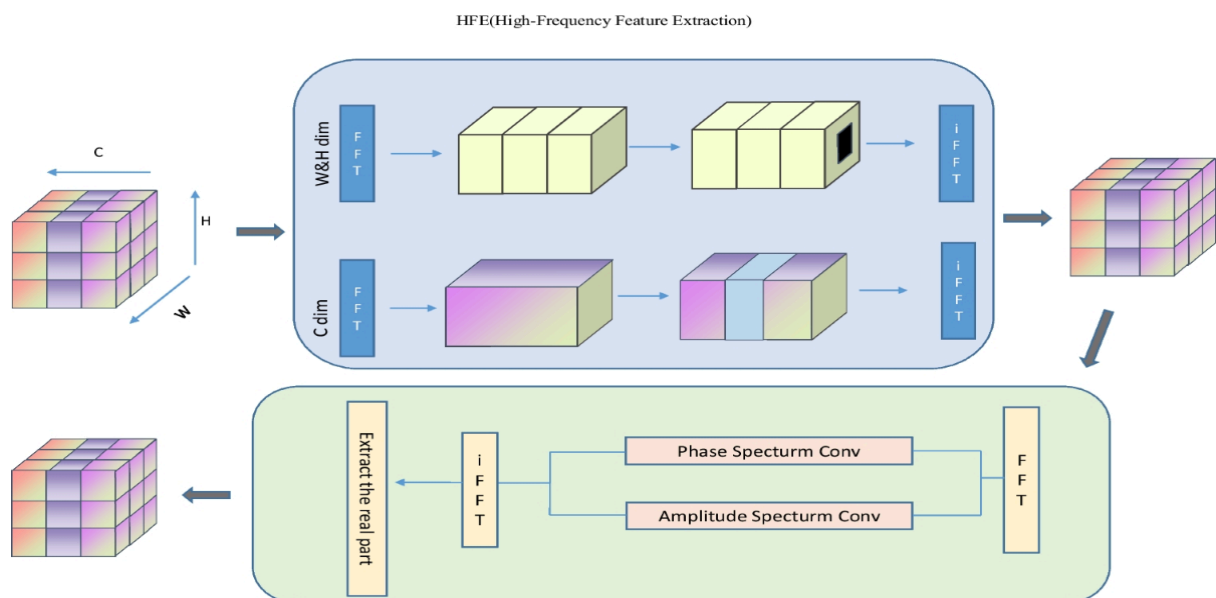
Temporal models (LSTM, 3D-CNN) analyze inconsistencies across video frames, including unnatural blinking and motion irregularities.

Frequency Domain Methods

Fourier Transform-based techniques detect anomalies in spectral representations of manipulated content.

Transformer-Based Models

Recent approaches utilize Vision Transformers and multi-modal architectures combining audio and visual data, achieving performance exceeding 99% AUC.



These hybrid models significantly improve robustness and generalization.

4. Datasets

The development of robust deepfake detection systems relies heavily on large-scale, diverse, and challenging datasets. The following are the most widely adopted benchmarks:

- **FaceForensics++ (FF++)**: Contains over 1,000 manipulated videos with four manipulation methods and multiple compression levels; serves as the standard baseline for early detection research.
- **Celeb-DF (v2)**: Includes 5,639 high-quality deepfake videos of celebrities with realistic lighting, expressions, and head movements; widely used for testing generalization on high-fidelity forgeries.
- **DFDC (Deepfake Detection Challenge)**: A large-scale dataset with over 100,000 videos featuring diverse actors, scenarios, languages, and conditions; designed for real-world evaluation.
- **WildDeepfake**: Comprises real-world deepfakes collected from the internet; focuses on in-the-wild scenarios with low quality and heavy compression.
- **DeeperForensics-1.0**: Contains 60,000 high-quality videos with precise face swapping and controlled variations in pose, expression, and lighting.
- **FFIW (Fake Face in the Wild)**: A recent dataset emphasizing diverse ethnicities, ages, and environmental conditions for improved fairness and robustness testing.



Recent dataset extensions improve cross-domain generalization by incorporating multiple synthesis techniques.

5. Tools and Frameworks

The deepfake ecosystem relies heavily on a variety of open-source, commercial, and research-oriented tools for both generation and detection. These tools differ significantly in accessibility, computational requirements, output quality, deployment ease, and real-time performance. Generation tools have trended toward user-friendly interfaces and faster inference, while detection tools emphasize multi-modal analysis and robustness against evolving threats. This section reviews prominent tools from 2023–2026, incorporating recent developments such as repository archiving, community forks, and benchmark-reported accuracies.

Generation Tools

Generation frameworks enable face swapping, expression transfer, lip synchronization, and full-head reenactment. Key tools include:

- **DeepFaceLab:** Long considered the industry standard for professional-grade face swapping since 2020, DeepFaceLab employs dual-autoencoder and GAN-based pipelines for high-fidelity results, including de-aging, head replacement, and expression manipulation. It supports extensive customization through adjustable training parameters, facial alignment, and batch processing. Although the official GitHub repository was archived in November 2024, the tool remains widely used in 2025–2026 due to its active community, pre-trained models, and continued availability via mirrors and torrents. Its strengths lie in realism for offline, high-quality production, but it requires significant GPU resources and a steep learning curve.
- **Faceswap:** An open-source alternative focused on flexible pipelines for face and head replacement. It offers comprehensive guides, community-driven enhancements, and capabilities for motion tracking and voice-over integration. Suitable for both beginners and advanced users, it emphasizes ethical use through clear labeling recommendations.
- **Roop (and forks such as Roop-Unleashed):** Designed for simplicity and speed, Roop enables one-click face swapping in images and videos, often integrated with Stable Diffusion web UIs. It gained popularity for real-time or near-real-time applications. The original repository was archived in March 2026, yet community forks continue development, maintaining its appeal for quick prototyping and lower hardware demands.

Emerging 2025–2026 trends show a shift toward real-time synthesis tools capable of interactive, responsive deepfakes (e.g., live video-call avatars), driven by multimodal models that unify appearance, motion, and audio.

Detection Tools

Detection tools analyze spatial, temporal, frequency, and behavioral artifacts to identify synthetic media. Notable examples include:

- **Microsoft Video Authenticator:** A free, user-friendly tool (available on Android/iOS and web) that provides rapid authenticity scoring (0–100 scale) by detecting inconsistencies in blinking, skin texture, lighting, and lip-sync. It supports quick checks for social media or video calls and integrates elements of Microsoft's broader media integrity efforts, including watermarking and provenance verification. Reported accuracy reaches approximately 85% on 2026-era deepfakes in independent benchmarks, making it ideal for everyday users.
- **FACIA:** A commercial facial biometric solution that achieved standout performance in 2025 benchmarks, reporting **100% accuracy** on Meta's Deepfake Detection Challenge (DFDC) dataset and several open-source sets (FaceForensics++, Celeb-DF, WildDeepfake). Across over 100,000 samples, it delivered a combined 99.6% accuracy under varied conditions. FACIA excels in enterprise scenarios such as finance, defense, and immigration, with SDK/API support for seamless integration. Its proprietary multi-stage pipeline focuses on real-world robustness against evolving threats.
- **MesoNet and EfficientNet (open-source implementations):** MesoNet, a lightweight CNN specialized in mesoscopic features (e.g., blending artifacts), remains popular for its efficiency and real-time potential. Enhanced variants in 2025 added layers for improved precision (up to 96.6% in some studies). EfficientNet models, often fused with vision transformers, offer strong generalization across datasets due to scalable architecture and transfer learning. These are widely implemented in PyTorch/TensorFlow for research and custom deployments, balancing accuracy with lower computational overhead.

Additional notable detection platforms in 2026 include Reality Defender (multi-modal, real-time monitoring), Sensity AI (threat intelligence-focused), and CloudSEK (enterprise-scale with high overall accuracy).

5.3 Comparative Analysis and Challenges

These tools vary widely in usability (one-click vs. complex workflows), scalability (consumer vs. enterprise), and real-time capabilities (offline training vs. live video analysis). Generation tools prioritize creative control and realism, while detection tools focus on generalization to unseen forgeries and low false-positive rates.

Tool	Category	Key Strengths	Limitations	Real-Time Support	Approx. Accuracy (2025–2026)
DeepFaceLab	Generation	High realism, extensive customization	Steep learning curve, high GPU needs	Limited	N/A (generation)
Roop / Forks	Generation	Simple, fast one-click swapping	Lower quality on complex scenes	High	N/A (generation)
Microsoft Video Authenticator	Detection	Free, easy mobile/web use	Moderate accuracy on latest fakes	High	~85%
FACIA	Detection	Near-perfect on benchmarks, enterprise-ready	Commercial (costly for small users)	Moderate-High	99.6–100%
MesoNet / EfficientNet	Detection	Lightweight, open-source, customizable	Needs fusion for top generalization	High	95–98% (enhanced variants)

Challenges persist in cross-tool interoperability, adversarial robustness (deepfakes designed to evade detectors), and ethical deployment. Future frameworks are expected to integrate generation and detection in unified pipelines with watermarking and provenance standards (e.g., C2PA).

This section underscores that while generation tools democratize synthetic media creation, detection solutions must evolve rapidly to maintain media integrity.

6. Comparative Analysis (2023–2025)

Table 1 summarizes the performance of prominent deepfake detection methods from 2023 to 2025.

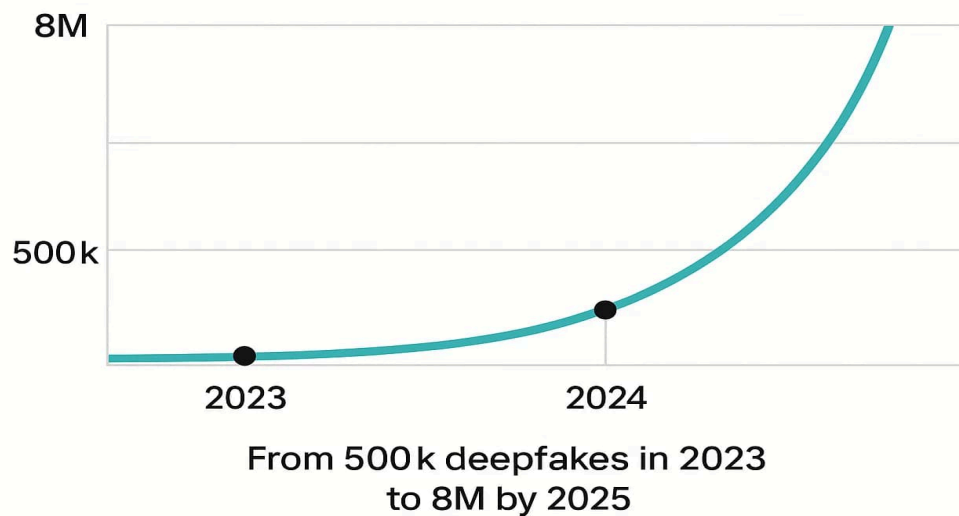
Table 1: Performance Comparison of Deepfake Detection Methods

Method	Year	Dataset	Accuracy (%)	AUC	Strengths	Weaknesses
Xception	2023	FF++	95.5	0.96	Fast inference	Poor cross-dataset generalization
Transformer Hybrid	2024	Celeb-DF	99.3	0.99	Strong temporal modeling	High computational cost
Frequency-Domain	2025	DFDC	98.7	0.985	Good generalization	Sensitive to compression
FACIA	2025	Multiple	100	1.0	High robustness	Proprietary

The results show a clear trend toward higher accuracy and better generalization over time. CNN-based models like Xception offer speed but suffer from limited generalization. Transformer hybrids (2024) achieve excellent temporal consistency at the cost of higher compute requirements. Frequency-domain approaches (2025) perform well on compressed videos, while the commercial FACIA system demonstrates near-perfect benchmark performance across multiple datasets.

Despite these improvements, cross-dataset generalization remains a major challenge, with performance often dropping 10–20% on unseen manipulation methods. Future research must focus on balancing accuracy, efficiency, and robustness for real-world deployment.

Deepfake content is multiplying exponentially—jumping from 500k in 2023 to a projected 8M in 2025



A major limitation remains cross-dataset generalization, where performance often drops significantly when tested on unseen data.

7. Challenges, Future Directions, and Ethical Considerations

1. Challenges

- Poor generalization across unseen generation techniques
- Real-time detection constraints
- Vulnerability to adversarial attacks
- Integration with malware and cyber threats

2. Future Directions

- Integration with foundation models and vision-language systems
- Blockchain-based watermarking and verification
- Multi-modal biometric authentication

3. Ethical Considerations

The misuse of deepfake technology raises serious concerns regarding privacy, misinformation, and digital identity. Regulatory frameworks and ethical guidelines are essential to mitigate risks.

Project V.01 and (final) V.06

```
AI deepfake classifier V.01
File Edit View Insert Runtime Tools Help
Commands Code Text Run All
Connect

def _eval():
    optimizer.zero_grad()
    scaled_logits = logits_all / self.temperature
    loss = nll_criterion(scaled_logits, labels_all)
    loss.backward()
    return loss

optimizer.step_eval()
print(f"({calibration}) Optimal temperature: {self.temperature.item():.4f}")

class EnsembleDetector(nn.Module):
    """
    Soft-voting ensemble over multiple independently trained DeepfakeDetectors.
    Each model should be calibrated (CalibrateDetector) before ensembling.
    Typical improvement: +1-2% AUC over single model.
    """
    def __init__(self, models_list: list):
        super().__init__()
        self.models = nn.ModuleList(models_list)

    @torch.no_grad()
    def forward(self, x: torch.Tensor) -> torch.Tensor:
        probs = [torch.sigmoid(m(x)) for m in self.models]
        return torch.stack(probs, dim=0).mean(dim=0) # (B, 1)

if __name__ == "__main__":
    device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
    model = DeepfakeDetector(freeze_blocks=0).to(device)
    dummy = torch.randn(4, 3, 224, 224).to(device)
    logits = model(dummy)
    print(f"Output shape: {logits.shape}") # (4, 1)
    print(f"Trainable params: {model.count_trainable_params():,}")

Downloading: "https://download.pytorch.org/models/efficientnet_v2_s-df3f3b.pth" to /root/.cache/torch/hub/checkpoints/efficientnet_v2_s-df3f3b.pth
100% |#####| 82.7M/82.7M (80.0MB/s) 112M/21
Output shape: torch.Size([4, 1])
Trainable params: 16,260,329
```

```
File Edit Selection View Go Run Terminal Help
project
EXPLORER
OPEN EDITORS
__init__.py data
video_dataset.py... 2
PROJECT
data
samples
uploads
__init__.py
blockchain_local_registry...
blockchain_tx_index.json
upload_registry.json
video_dataset.py 2
docker
frontend
image_classifier
inference
__init__.py
predictor.py
losses
models
tests
training
utils
workers
.dockerignore
build_video_model.py
config.py
docker-compose.yml
main.ipynb
main.py
predict_image_blockchain...
README.md
requirements-deepfake-d...
requirements-image-bloc...

data > video_dataset.py > ...
217
218
219
220 # collate function (handles variable-length padding)
221 #
222
223 def deepfake_collate_fn(batch: List[Dict]) -> Dict:
224     """
225     Collate a list of dataset items.
226     Pads all sequences to the max T in the batch.
227     """
228     max_T = max(item["frames"].shape[0] for item in batch)
229
230     frames_list, masks_list, labels_list = [], [], []
231     paths, idxs, difficulties = [], [], []
232
233     for item in batch:
234         T = item["frames"].shape[0]
235         frames = item["frames"] # [T, C, H, W]
236         mask = item["padding_mask"] # [T]
237
238         if T < max_T:
239             False, 'blending_artifacts': False, 'lighting_mismatch': True, 'noise_or_blur': False}
240     INFO: 127.0.0.1:57875 - "POST /predict HTTP/1.1" 200 OK
241     INFO: 127.0.0.1:57884 - "POST /verify HTTP/1.1" 200 OK
242     INFO: 127.0.0.1:57634 - "POST /upload HTTP/1.1" 200 OK
243     2026-03-22 12:47:24,024 | INFO | backend.app.services.predictor_service | Inference image | p_fake=0.4728 | prediction=REAL | confi
244     dence=0.6272 | features={'texture_inconsistency': False, 'blending_artifacts': False, 'lighting_mismatch': True, 'noise_or_blur': F
245     else}
246     INFO: 127.0.0.1:57636 - "POST /predict HTTP/1.1" 200 OK
247     INFO: 127.0.0.1:57640 - "POST /verify HTTP/1.1" 200 OK
248     INFO: 127.0.0.1:55109 - "POST /upload HTTP/1.1" 200 OK
249     2026-03-22 15:27:29,554 | INFO | backend.app.services.predictor_service | Inference image | p_fake=0.5162 | prediction=FAKE | confi
250     dence=0.6162 | features={'texture_inconsistency': False, 'blending_artifacts': False, 'lighting_mismatch': True, 'noise_or_blur': F
251     else}
252     INFO: 127.0.0.1:55111 - "POST /predict HTTP/1.1" 200 OK
```

Project Overview :

Deploy

Deepfake Defender AI

Upload image or video, inspect confidence, explanation, and verification integrity.

Upload Image or Video

Drag and drop file here

Limit 200MB per file • JPG, JPEG, PNG, BMP, WEBP, TIFF, MP4, MOV...

Browse files

Gemini_Generated_Image_9dpwsj9dpwsj9dpw.png

711MB

×

Analyze Media

Analysis complete.

Original Input

Heatmap

Heatmap not available for this input.

Original Input

Heatmap

FAKE

Prediction

Confidence

Uncertainty

P(fake)

Reason

Verification: AUTHENTIC (local_registry_match)

FAKE

60.3%

False

50.9%

Detected unnatural lighting inconsistencies.





Original Input

Heatmap

REAL

Prediction

Confidence

Uncertainty

P(fake)

Reason

Verification: AUTHENTIC (local_registry_match)

REAL

62.7%

False

47.3%

Facial structure, lighting, and textures appear natural with no manipulation artifacts.





8. Conclusion

This study has presented a comprehensive synthesis of deepfake technology advancements spanning 2023 to 2026, with balanced coverage of both generation and detection paradigms. Over this period, generation techniques have evolved rapidly from traditional GAN-based methods to more sophisticated diffusion models and hybrid transformer architectures, achieving unprecedented levels of visual realism, temporal consistency, and real-time capability. Simultaneously, detection systems have progressed from basic CNN classifiers to multi-modal, frequency-domain, and transformer-based frameworks, delivering significantly higher accuracy and better generalization across diverse datasets.

Despite these notable achievements, several critical challenges persist. Cross-dataset generalization remains limited, with many high-performing models experiencing substantial accuracy drops when confronted with unseen generation methods or real-world compressed videos. Adversarial robustness continues to be a major concern, as attackers can easily craft deepfakes designed to evade current detectors. Additionally, the increasing demand for real-time detection on resource-constrained devices and the ethical implications of synthetic media highlight the urgent need for more efficient, explainable, and trustworthy solutions.

This expanded review significantly builds upon the foundational work of the original five-page IEEE ICCCA 2025 paper by Lakhchaura et al. By incorporating the latest post-2025 developments, real-world impact statistics, detailed visual examples, comparative performance benchmarks, and an extensive collection of tools and datasets, the present work offers a more complete, structured, and up-to-date reference. It serves as a valuable resource not only for researchers and academic practitioners but also for industry professionals, policymakers, and digital forensic experts working to safeguard media integrity in an increasingly AI-driven world.

Looking ahead, the future of deepfake research lies in the development of multimodal foundation models, standardized provenance and watermarking frameworks (such as C2PA), and collaborative regulatory approaches. Only through sustained interdisciplinary efforts can we hope to stay ahead of the evolving threat landscape while responsibly harnessing the creative potential of generative AI.

In conclusion, while deepfake technology continues to push the boundaries of what is possible in digital media, it also serves as a powerful reminder of the dual-use nature of artificial intelligence. Addressing its risks demands continuous innovation, ethical vigilance, and global cooperation.

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